

## Title & Team Information

**Title:** Crime and What Else? Investigating the Predictive Power of Social Factors on Incarceration Rates

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**Stakeholder POs:** Mary Rose Krouse, Shanti Brodnick, & Rohan Srinivas Babu

**Studio session:** 1

**Team size:** solo

## Problem Framing & Research Questions

**Problem statement:** The United States prison system currently holds nearly 2 million people, and for decades has imprisoned its inhabitants at one of the highest rates in the world (Lurigio, 2024). A variety of legal factors, including mandatory minimum sentences and “three strikes” laws, have contributed to the unprecedented rise in incarceration in the US since the 1970s and 80s (DeFina and Hannon, 2013; Lurigio, 2024). The US incarceration rate has since levelled off, remaining relatively steady since the 2010s; however, this does not reflect the nation’s crime rate, which has fallen since peaking in in the 1990s (Pettit and Gutierrez, 2013). Given that the US’s criminal legal system wields tremendous power—power that is most often turned against already disadvantaged individuals and their communities—it is important that we recognize protective and risk factors for incarceration, many of which are social in nature (Lurigio, 2024; Henry, 2020; Pettit and Gutierrez, 2013). This project seeks to interrogate the dominant narrative that crime is the primary cause of incarceration by examining three other risk factors.

### Research questions:

- How do the predictive power of social factors (such as poverty, race, or government spending on police) compare to that of violent and property crime rates when using support vector regression to model the national incarceration rate?
  - If crime is in fact the best predictor of incarceration rates, which social factor is the best predictor of crime rates?

The main change in my questions since writing the charter is that I no longer plan to look into recidivism rates, primarily because the data simply is not available in a manner that is consistent with the rest of the data I will be using. Instead, I have introduced a third social factor to interrogate, which is government spending on police. In anticipation of the possibility that crime does turn out to be the best predictor, I also aim to dig deeper and repeat the prediction process to examine which of the three social factors best predicts the crime rate.

## Impact & Stakeholders

The key stakeholders in this work are those who are most often targeted by criminalization and incarceration, primarily poor and BIPOC folks (Pettit and Gutierrez, 2013). Additionally,

communities across the country are continually harmed by increasing incarceration rates, as incarceration has been shown to negatively impact both community health (Wildeman and Wang, 2017) and poverty (DeFina and Hannon, 2013). By critically examining the circumstances that can contribute to incarceration, this project seeks to uplift the work of organizations such as Transformative Justice Community and the Oregon Justice Resource Center, which advocate for solutions to harm that shift our paradigm away from imprisonment.

My shared theme is *social justice and identity-driven disparity analysis*. As others who are interested in social justice but perhaps not as familiar with the prison-industrial complex as I am, I hope that they'll point out anywhere I make assumptions/lack context and help ensure that my project doesn't steer anyone toward misinformed conclusions.

## Data Sources & Access Plan

### *Corrections Statistical Analysis Tool: Prisoners*

- Year-end prison population counts by state, 1978-2021
- <https://csat.bjs.ojp.gov/advanced-query> (one-time csv download)
- Volume: 2,376 rows x 3 columns
- Approval status: requested

### *Number of Poor and Poverty Rate by State*

- Yearly counts and percentages of state poverty levels, 1959-2024
- <https://www.census.gov/data/tables/time-series/demo/income-poverty/historical-poverty-people.html> (one-time xlsx download)
- Volume: 2,350 rows x 6 columns
- Approval status: requested

### *Summary Reporting System: Estimated Crimes*

- Yearly estimated counts of 4 types of violent crimes and 3 types of property crimes (as well as aggregates) by state, 1979-2024
- <https://cde.ucr.cjis.gov/LATEST/webapp/#/pages/downloads> (one-time csv download)
- Volume: 2,389 rows x 15 columns
- Approval status: requested

### *IPUMS CPS*

- Household- and individual-level data collected yearly as part of the Current Population Survey's Annual Social and Economic Supplement (available 1962-2025). Includes the survey year, the race and state of residence of individuals in sampled households, and a weight to be used when calculating aggregates.
- <https://cps.ipums.org/cps-action/variables/group>
- Volume: 12,821,891 rows x 17 columns
- Approval status: requested

Spending on police (will be constructed from the sources below):

- *Survey of Government Finance Historical Data*
  - Yearly data on all state and local government spending, including state-level spending on police protection (per capita), 1951-2008
  - <https://www.census.gov/programs-surveys/gov-finances/data/historical-data.html> (one-time csv download)
  - Volume: 2,988 rows x 11 columns (pre-queried for police spending specifically)
  - Approval status: requested
- *State Government Finances Individual Unit Data*
  - Yearly data on all state and local government spending, including state-level spending on police protection (absolute), 2012-2024
  - [https://data.census.gov/table/GOVSSTATEFINTIMESERIES.GS00STATEFIN03?q=State%20and%20Local%20Government%20Finances&nkd=AGG\\_DESC~SF0158%3ASF0342%3ASF0343%3ASF0344%3ASF0345%3ASF0346%3ASF0365&colSort=AGG\\_DESC\\_LABEL~desc](https://data.census.gov/table/GOVSSTATEFINTIMESERIES.GS00STATEFIN03?q=State%20and%20Local%20Government%20Finances&nkd=AGG_DESC~SF0158%3ASF0342%3ASF0343%3ASF0344%3ASF0345%3ASF0346%3ASF0365&colSort=AGG_DESC_LABEL~desc) (one-time csv download)
  - Volume: 3,520 rows x 9 columns (pre-filtered for police spending specifically)
  - Approval status: requested
- *State Government Finances: Individual years*
  - Data for specific years on all state and local government spending, including state-level spending on police protection (absolute), for the years not covered by the datasets above (2009-2011)
  - <https://www.census.gov/data/tables/2009/econ/state/historical-tables.html> (2009), <https://www.census.gov/data/tables/2010/econ/state/historical-tables.html> (2010), <https://www.census.gov/data/tables/2011/econ/state/historical-tables.html> (2011), all one-time downloads
  - Volume: 64 rows x 52 columns for all
  - Approval status: requested

## Data Engineering Plan

Because this is historical data, all data sets will be ingested as either .csv or .xlsx files via one-time download, depending on which format is available. Data will be stored in the project repo on Github (outlined below), and as processing progresses updated versions of each dataset will be uploaded to the interim or processed folders as necessary.

The current structure of the project's data folder, based on Professor Cordova's template, is:

- `../data/raw`: contains all original one-time data downloads (incarceration, crime, and poverty)
  - Note: The raw IPUMS CPS data will not be present in this folder, as it can be read directly into R using an IPUMS package and when converted into a csv is too large to be stored in Github.
- `../data/raw/spending`: contains all the raw files that will be used to construct a full police spending dataset

- `../data/interim`: will hold the individual datasets once they have each been cleaned (filtered, aggregated, and/or used to calculate per capita values as needed)
- `../data/processed`: will contain the final dataset for use with SVR, consisting of the relevant columns from each cleaned dataset combined into one file.

## Planned Analysis

The key method of this project will be support vector regression (SVR), chosen because of its ability to deal with complex, non-linear data while remaining more straightforward than methods such as XGBoost. Save for an adjustment to the kernel (which will be either poly or rbf), I will use the default SVR model across all variables to ensure they are being treated the same. My focus in this project will be on predicting the *national* incarceration rate based on each states' chosen measures; I will plan to train one SVR model for each predictive factor to ensure that no other data will affect predictions.

Three evaluation metrics will be used: correlation coefficient, r2 score, and root mean squared error (RMSE). The correlation coefficient will be used as a preliminary evaluation of the linearity of relationships between my variables, with the expectation that none will be particularly linear. Once all SVR models have been trained and used to create incarceration rate predictions based on test data stratified by state, the r2 score will be used to evaluate how much of the variance in incarceration rate can be explained by the model, while the RMSE will measure the average error across predictions. My research questions will be answered when these three measures have been calculated for all applicable variables/models and I can compare performance across the models.

## Program Integration

- **Data engineering**: Raw, in-process, and cleaned data will be separated to ensure data hygiene. Data from multiple disparate sources will be combined based on shared characteristics to create a final dataset to work from.
- **Analytics/ML**: As noted above, this project hinges on machine learning. The primary mode of analysis will be evaluating the accuracy of incarceration rate predictions made by a support vector regression algorithm based on the factors I have outlined.
- **Visualization and communication**: A key focus of this project will be making the results of the support vector regression algorithm interpretable and legible to a wider audience. That will require finding an accessible way to communicate very technical information, which will include using graphics to display evaluation metrics such as the accuracy of predictions.
- **Ethics and responsible use**: As outlined below, I will focus on clearly communicating the limitations and observations of this project in an effort to avoid the possibility of any findings being weaponized against the mission and vision of the project.

- **Statistics/research design:** Combining 9 different data sets, each with their own quirks, will necessitate understanding how to work with data that have been collected in a variety of formats. For example, aggregating the IPUMS CPS data will require the incorporation of weighting, since these data are in a survey format; among the aggregated data sets, measures will have to be standardized in order to make them comparable across both geography and time.

## Ethics & Risk Notes

Most of the data I will be using are not particularly risky in and of themselves, as all of it is highly aggregated to the state and year level; thus privacy of individuals is protected. The only individual-level data source I will be using is the IPUMS Current Population Count Survey, which has been pre-anonymized (individuals are known only by their household ID and personal number) and will be further aggregated up to the state level in order to match the rest of my sources. To that end, one of the biggest risks of this project is the possibility of my findings being misconstrued and/or weaponized against those who have the potential to benefit most from this project (e.g. by characterizing all poor people as criminals). In order to combat that risk I will have to be very clear about the limitations of this project, both in terms of the data and in terms of its nature as a correlation-based analysis.

One major limitation is present in the nature of crime data: because it relies only on crimes reported, the data likely is not an accurate reflection of the actual crime rate (and in some cases specific states in certain years have their own caveats). Further, it must be noted that no social factor exists in isolation, and in reality many variables interact in complex ways that can contribute to or protect against risk of criminalization and incarceration. The nature of this project as a solo, single-semester endeavor means that I cannot possibly account for all the possible factors that impact incarceration; rather, I seek to gain a deeper understanding of just a few of the factors that may have an outsize impact on the course of individuals' lives.

## Timeline & Milestones

| Week   | Milestone           | Tasks                                                                                                                                                       |
|--------|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 4 | M1-project-proposal | <ul style="list-style-type: none"> <li>● Source data</li> <li>● Finalize research question, methods</li> </ul>                                              |
| Week 5 |                     | <ul style="list-style-type: none"> <li>● Come up with detailed cleaning plan</li> <li>● Clean incarceration and poverty data according to plan</li> </ul>   |
| Week 6 |                     | <ul style="list-style-type: none"> <li>● Finish cleaning, process, and combine data</li> <li>● Pull summary statistics, EDA, preliminary metrics</li> </ul> |

|         |                  |                                                                                                                                                               |
|---------|------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Week 7  | M2-data-summary  | <ul style="list-style-type: none"> <li>• Write up milestone &amp; documentation</li> <li>• Write repeatable pseudocode that can be utilized for ML</li> </ul> |
| Week 8  |                  | <ul style="list-style-type: none"> <li>• Start running ML</li> <li>• Plan visualizations</li> </ul>                                                           |
| Week 9  |                  | <ul style="list-style-type: none"> <li>• Finish running, evaluate ML</li> <li>• Draft visualizations</li> </ul>                                               |
| Week 10 | M3-poster-draft  | <ul style="list-style-type: none"> <li>• Write up important points, highlights</li> <li>• Preliminary poster formatting</li> </ul>                            |
| Week 11 |                  | <ul style="list-style-type: none"> <li>• Outline structure of writeup, start writing</li> <li>• Continue refining poster</li> </ul>                           |
| Week 12 | M4-writeup-draft | <ul style="list-style-type: none"> <li>• Complete draft writeup</li> <li>• Continue refining poster</li> </ul>                                                |
| Week 13 |                  | <ul style="list-style-type: none"> <li>• Finish poster</li> <li>• Edit writeup</li> </ul>                                                                     |
| Week 14 | M5-final         | <ul style="list-style-type: none"> <li>• Finish!</li> </ul>                                                                                                   |

## DS3 Process & Tooling

- Github repo: <https://github.com/eskerani/prison-predictor-capstone/tree/main>
- Github project: <https://github.com/users/eskerani/projects/1>
- Discord category: Project 12: Emery-\*. Channels include #general, #standup, #studio, and #blockers. I am in my own project category, as well as those of the peer projects I will be reviewing.
- The charter has been filled out and committed; PBIs have been added to the README and project board.
- I will aim to post iteration reviews by the Saturday night before class so that peers have adequate time to access them. Once briefs have been posted (by midnight the Sunday before class), I will acknowledge them in the studio discord channel by either messaging or reacting within 24 hours, as well as making preliminary decisions about which suggestions will be accepted, declined, or put on hold. Similarly, I will acknowledge critiques (due by midnight the Tuesday after class) within 24 hours of posting and will subsequently update PBIs with any follow-up items.

## References

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